Granular Instrumental Variables

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The Law of Large Numbers fails in modern economies

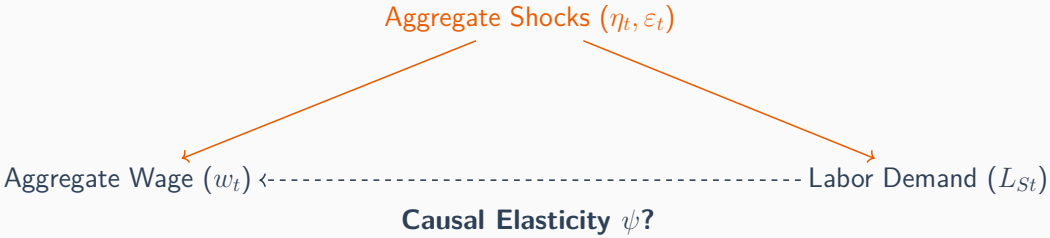
In a standard economy, micro-shocks cancel out at a rate of $1/\sqrt{N}$ Proof.

In a **granular** economy, where the size distribution of firms follows a power law, a few firms are so large that their idiosyncratic luck drives aggregate volatility.

Endogenous aggregate shocks confound the effect of large firms

We want to estimate the elasticity of wages to labor demand.

The fundamental challenge:



The Simple Model

Difference-in-averages as an instrument

A simple model of aggregate and idiosyncratic shocks

Consider an economy with N firms and a single common factor (η_t).

The system is given by:

$$w_t = \psi L_{St} + \varepsilon_t \tag{Supply}$$

$$L_{it} = \phi^d w_t + \eta_t + u_{it} \tag{Firm Demand}$$

Where w_t is the wage, $L_{St} = \sum_i S_i L_{it}$ is the size-weighted aggregate outcome (number of workers), η_t and ε_t are aggregate shocks, and u_{it} is the idiosyncratic shock to firm i.

OLS fails because aggregate wages and common shocks are correlated

We cannot estimate ψ and ϕ^d by OLS.

Why?

- ε_t and η_t are typically correlated.
- This implies that L_{St} is correlated with ε_t in the supply equation (Supply).
- And w_t is correlated with η_t in the demand equation (Firm Demand).

Constructing the Granular Instrumental Variable (z_t)

The GIV exploits the difference between size-weighted and equal-weighted averages.

Define the equal-weighted average outcome:

$$L_{Et} = \frac{1}{N} \sum_{i=1}^{N} L_{it} \qquad \text{(Equal-Weighted Average)}$$

The GIV is the size-weighted outcome minus the equal-weighted outcome:

$$z_t := L_{\Gamma t} = L_{St} - L_{Et} \qquad \text{(GIV Definition)}$$

The GIV isolates purely idiosyncratic firm shocks

Substitute the demand equation into the aggregate definitions:

$$L_{St} = \phi^d w_t + \eta_t + u_{St}$$
$$L_{Et} = \phi^d w_t + \eta_t + u_{Et} \qquad \text{(Aggregate Demands)}$$

The common components $(\phi^d w_t + \eta_t)$ difference out:

$$z_t = u_{\Gamma t} = u_{St} - u_{Et} = \underbrace{\sum S_i L_{it}}_{\text{Size-Weighted}} - \underbrace{\frac{1}{N}\sum L_{it}}_{\text{Equal-Weighted}} \qquad \text{(Isolated Shocks)}$$

z_t is made *only* of idiosyncratic shocks.

The GIV yields both supply and demand elasticities

By exogeneity, $\mathbb{E}[u_t \varepsilon_t] = 0 \implies \mathbb{E}[z_t \varepsilon_t] = 0$.

Using the supply equation (Supply), $\mathbb{E}[(w_t - \psi L_{St}) z_t] = 0$:

$$\psi = \frac{\mathbb{E}[w_t z_t]}{\mathbb{E}[L_{St} z_t]} \qquad \text{(Supply Elasticity)}$$

Using the equal-weighted demand equation from (Aggregate Demands), $\mathbb{E}[(L_{Et} - \phi^d w_t) z_t] = 0$:

$$\phi^d = \frac{\mathbb{E}[L_{Et} z_t]}{\mathbb{E}[w_t z_t]} \qquad \text{(Demand Elasticity)}$$

The Full Model

Handling heterogeneous factor loadings

Superstar firms respond differently to macro trends

The simple GIV $(L_{St} - L_{Et})$ assumes every firm has the same loading $\lambda_i = 1$. If large firms are more sensitive to the Fed, the simple GIV is **"leaky"**—it correlates with η_t.

The Full Factor Model for granular firms

We now allow for heterogeneous loadings λ_i:

$$L_{it} = \phi w_t + \lambda_i \eta_t + u_{it} \qquad \text{(Heterogeneous Loadings)}$$

The simple GIV becomes:

$$z_t = (\lambda_S - \lambda_E)\eta_t + (u_{St} - u_{Et}) \qquad \text{(Leaky GIV)}$$

If $\lambda_S \neq \lambda_E$, the instrument is **invalid** ($\mathbb{E}[z_t \eta_t] \neq 0$).

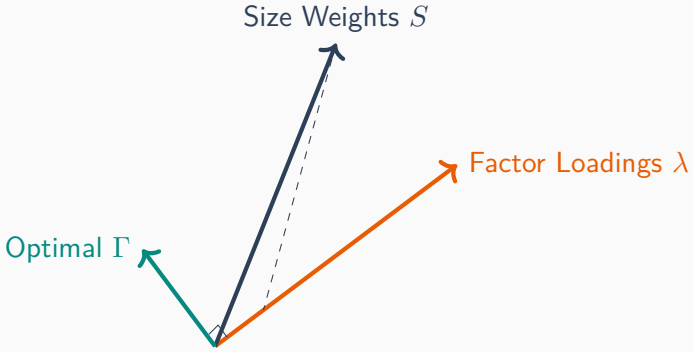
The Solution: Orthogonal Optimal Weights Γ

We need weights Γ such that the factor components **cancel exactly**.

$$\sum_{i=1}^{N} \Gamma_i \lambda_i = 0 \quad \text{(Orthogonality Condition)} \qquad \text{(Orthogonality)}$$

This ensures $z_t = \sum \Gamma_i u_{it}$ is purely idiosyncratic.

Geometry of GIV: Projecting away from the macro trend



Γ is the component of S orthogonal to λ.

Implementation and Assumptions

Identifying wage spillovers from superstar firms

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How do we apply GIVs to labor markets?

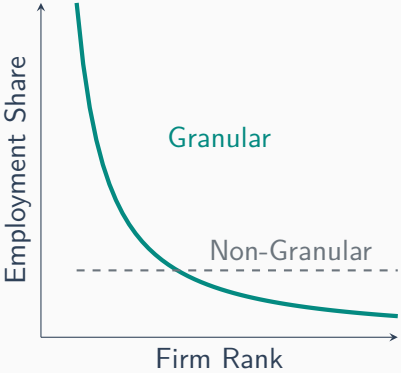
- **The granular shock:** Use idiosyncratic productivity or demand shocks of the largest employers in a region.
- **The instrument:** The GIV (z_t) aggregates these large-firm idiosyncratic shocks.
- **The estimation:** Instrument local or aggregate wage changes (w_t) using the GIV to estimate the causal impact of labor demand on wages (ψ).

GIV precision relies on concentrated labor markets and volatile micro-shocks

For the instrument to be strong, we need:

1. **A large excess Herfindahl index (h):** A few superstar firms must account for a substantial fraction of total employment.

2. **Volatile idiosyncratic shocks (σ_u):** Firm-specific shocks must be large relative to aggregate macroeconomic volatility (σ_ε).



Appendix

Aggregate volatility decays at $1/\sqrt{N}$ under equal weights

Suppose there are N firms, each receiving an independent idiosyncratic shock u_{it}. Assume these shocks have mean 0 and a common variance σ^2:

$$\mathbb{E}[u_{it}] = 0, \quad \text{Var}(u_{it}) = \sigma^2, \quad \text{Cov}(u_{it}, u_{jt}) = 0 \text{ for } i \neq j \qquad \text{(Shock Assumptions)}$$

In a non-granular economy, the aggregate shock is an **equal-weighted** average:

$$u_{Et} = \frac{1}{N} \sum_{i=1}^{N} u_{it} \qquad \text{(Equal-Weighted Shock)}$$

The variance of the aggregate shock is:

$$\text{Var}(u_{Et}) = \text{Var}\left(\frac{1}{N} \sum u_{it}\right) = \frac{1}{N^2} \sum \text{Var}(u_{it}) = \frac{N\sigma^2}{N^2} = \frac{\sigma^2}{N} \qquad \text{(Variance Decay)}$$

The aggregate volatility is therefore: $\sigma_{agg} = \frac{\sigma}{\sqrt{N}}$